

Connection Oriented and Connectionless Services

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Connection-Oriented Service

- A connection-oriented protocol establishes a dedicated, end-to-end connection between the sender and the receiver before transmitting data.
- This ensures both ordered and reliable delivery of packets, making it highly suitable for applications where accuracy and sequence matter.

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- Core concepts specific to connection-oriented protocols include:
 - Connection Establishment: A “handshake” process initiates and establishes a connection before data transmission.
 - Ordered Delivery: Packets are transmitted sequentially, ensuring they are received in the correct order.
 - Reliable Delivery: Acknowledgments and retransmissions ensure all packets arrive intact and uncorrupted.
 - Virtual Circuit: A logical pathway is created, mimicking a direct physical link throughout the session.
 - Connection Overhead: The handshake process and acknowledgment mechanisms introduce additional overhead.
 - Higher Latency: The focus on reliability often results in slightly higher delays.

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- The Transmission Control Protocol (TCP) is the most well-known example of a connection-oriented protocol:
 - Three-Way Handshake: TCP initiates a connection with a three-step process involving SYN (synchronize), SYN-ACK (synchronize acknowledgment), and ACK (acknowledgment). This ensures both sender and receiver are ready to exchange data.
 - Data Transfer with Sequencing and Acknowledgments: Data is split into packets, each numbered sequentially. The receiver acknowledges the arrival of packets, ensuring none are lost or out of order.
 - Retransmissions: If any packet is lost, the protocol automatically retransmits it until it is successfully delivered.
 - Connection Termination: Once the transfer is complete, TCP gracefully closes the connection through a four-step handshake, ensuring no data is left behind.

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Transmission Control Protocol - TCP

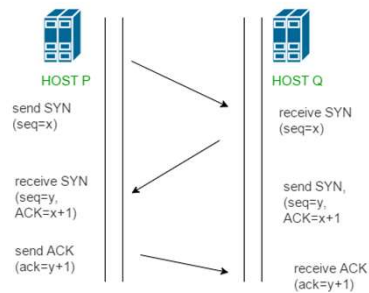
- TCP establishes a reliable connection between sender and receiver using the **three-way handshake (SYN, SYN-ACK, ACK)** and it uses a **four-step handshake (FIN, ACK, FIN, ACK)** to close connections properly.
- It ensures **error-free, in-order delivery** of data packets.
- It uses **acknowledgments (ACKs)** to confirm receipt.
- It prevents data overflow by adjusting the data transmission rate according to the receiver's buffer size.
- It prevents network congestion using algorithms like **Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery**.
- TCP header uses **checksum** to detect corrupted data and requests retransmission if needed.
- It is used in applications requiring **reliable** and **ordered** data transfer, such as web browsing, email, and remote login.

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- **Working of TCP**

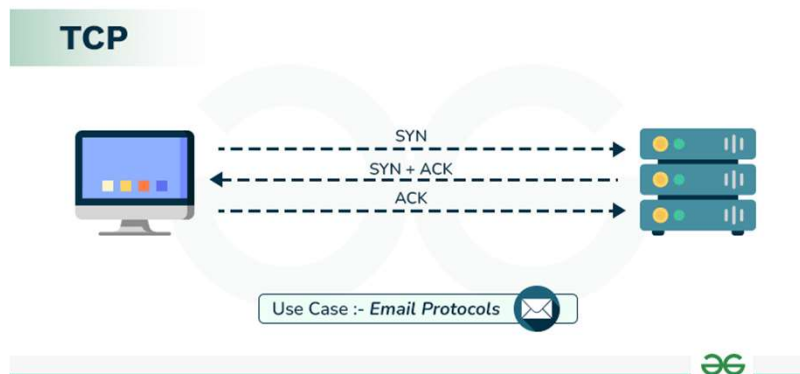
- **Data Segmentation:** The application data is divided into small segments for efficient transmission instead of sending it all at once.
- **Routing:** Each segment can take a different route through the network, depending on traffic or congestion, but all segments eventually reach the same destination.
- **Reassembly:** At the receiver's end, TCP reassembles these segments into the original message using **sequence numbers**.
- **Acknowledgment:** The receiver sends **ACK (acknowledgment)** messages back to confirm successful receipt of each segment. If an acknowledgment isn't received, TCP retransmits the missing segment.
- **Error and Flow Control:** TCP uses **checksums** to detect errors and a **sliding window mechanism** to control the flow of data, ensuring the receiver isn't overwhelmed.

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- **Step 1 (SYN):** In the first step, the client wants to establish a connection with a server, so it sends a segment with SYN(Synchronize Sequence Number) which informs the server that the client is likely to start communication and with what sequence number it starts segments with
- **Step 2 (SYN + ACK):** Server responds to the client request with SYN-ACK signal bits set. Acknowledgement(ACK) signifies the response of the segment it received and SYN signifies with what sequence number it is likely to start the segments with
- **Step 3 (ACK):** In the final part client acknowledges the response of the server and they both establish a reliable connection with which they will start the actual data transfer.

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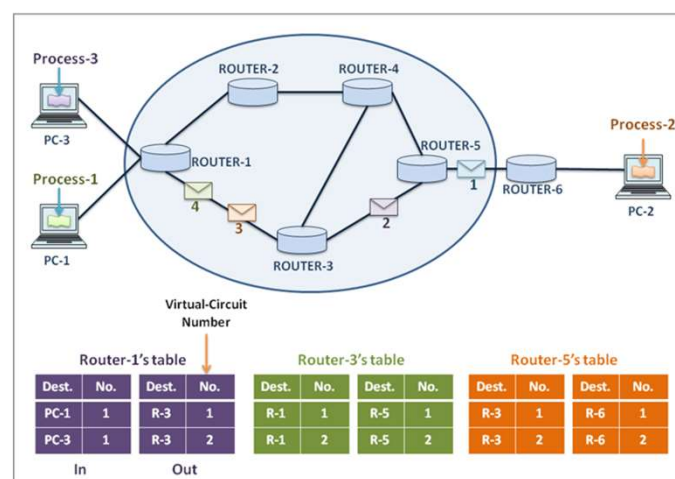
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- **Features of TCP**

- **Segment Numbering:** Each byte of data is numbered, and segments carry sequence numbers. Acknowledgment numbers confirm successful receipt.
- **Connection-Oriented:** Sender and receiver remain connected until data transfer is complete. Data order is preserved.
- **Full Duplex:** Data can flow in both directions simultaneously, improving efficiency.
- **Flow Control:** Uses a **sliding window** to prevent the sender from overwhelming the receiver.
- **Error Control:** Detects and manages corrupted, lost, duplicate, or out-of-order segments to ensure reliability.
- **Congestion Control:** Adjusts the sending rate based on network congestion to avoid overload.

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Implementation of Virtual Circuit Networks



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- As shown in the figure, the connection is established before the communication between PC-1 and PC-2.
- PC-1 is sending four packets to PC-2. Here, the identifier -1 is assigned to the packets of PC-1. Packet-1, Packet-2, Packet-3 and Packet-4 are transmitted to PC-2 with the identifier -1.
- Now, PC-3 also wants to send a packet to PC-2. It asks for identifier-1 from the network layer to establish a virtual-circuit network. But it is already assigned to PC-2. The router-1 assigns identifier-2 to PC-3.
- In the end, the routing table has two entries, one for PC-1 and the other for PC-3.
- Different identifiers are used to avoid packet collisions on the network.

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Connectionless Service

- A connectionless protocol transmits data without establishing a dedicated connection. Each data packet, also known as a datagram, is sent independently and contains all the necessary destination information. These packets may follow different routes and do not guarantee reliable or ordered delivery.

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- Core concepts specific to connectionless protocols include:
 - No Connection Setup: Data transmission begins immediately without a setup phase.
 - Independent Packets: Each packet is routed independently, which can lead to reordering in transit.
 - Best-Effort Delivery: Packets are sent without guarantees; some might be lost, duplicated, or received out of order.
 - Lower Overhead: Minimal processing is required, resulting in faster data transmission.
 - Faster Transmission: The absence of confirmations and retransmissions allows for near-real-time performance.
 - Unreliable Delivery: There's no guarantee that all packets will reach the destination.
 - No Inherent Ordering: Packets may arrive out of sequence, requiring additional mechanisms if order matters.

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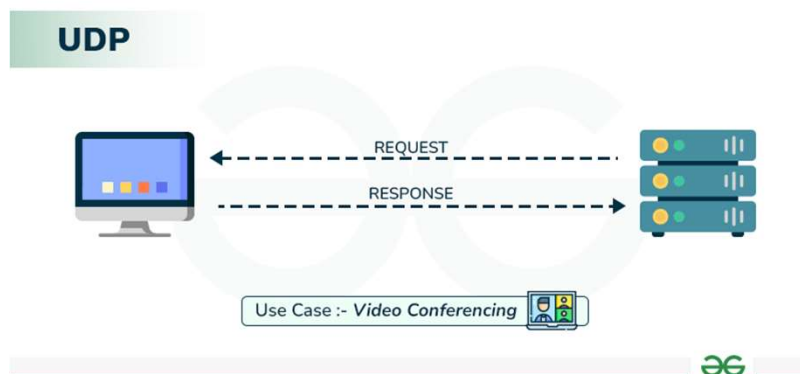
- The User Datagram Protocol (UDP) exemplifies the connectionless approach:
 - Packet Encapsulation with Destination Address: Each UDP packet, or datagram, includes the destination IP address and port number, enabling it to reach the intended recipient.
 - Independent Routing: Packets are routed independently. They may take different paths and arrive out of order or not at all.
 - No Acknowledgments or Retransmissions: UDP makes no effort to confirm packet delivery, minimizing processing time and maximizing speed.

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User Datagram Protocol - UDP

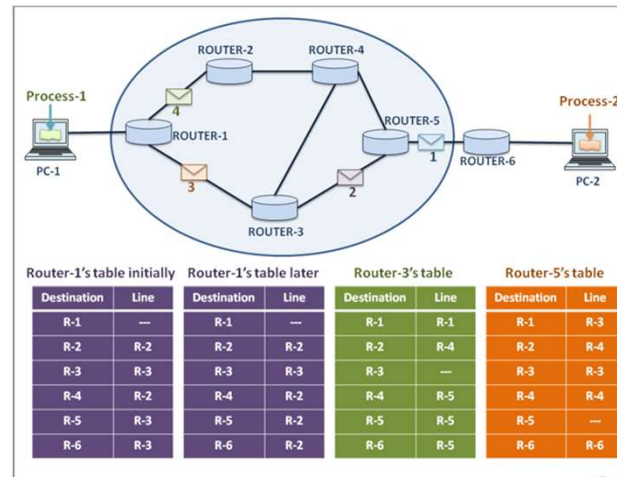
- User Datagram Protocol (UDP) is a Transport Layer protocol of the Internet Protocol (IP) that provides fast, connectionless, and lightweight communication between processes. It does not guarantee delivery, order, or error checking, making it suitable for real-time and time-sensitive applications such as video streaming, DNS, and VoIP.

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Implementation of Datagram Networks



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- As shown in the figure, PC-1 and PC-2 want to communicate. PC-1 needs to send the packets to PC-2. Process-1 and Process-2 are running on PC-1 and PC-2, respectively.
- The network layer communicates with the transport layer and sends the message with instructions to deliver packets to the PC-2 process.
- The network layer breaks the packet into four packets, packet-1, packet-2, packet-3, and packet-4.
- All four packets are transmitted to Router-1 using the point-to-point protocol. Router-1 has an initial router table that decides where to send the packets.
- Each router has two columns, Destination, and Line. Using the line column, the router determines the path to reach the other router. But, it uses only directly connected lines.
- For example, Router-1 has two outgoing lines Router-2 and Router-3. So Router-1 will send each packet to Router-2 and Router-3.
- Packet-1, Packet-2 and Packet-3 are transmitted through the path Router-1 -> Router-3 -> Router-5 -> Router-6 -> PC-2, while Packet-4 is transmitted through the path Router-1 -> Router-2 -> Router-4 -> Router-5 -> Router-6 -> PC-2.
- But why is packet-4 traveling on a different path than the path of the first three packets? Because Router-1 understood the network and detected a traffic jam somewhere along the 1-3-5-6-PC2 way.
- The routing algorithm is used to manage and make routing decisions based on a routing table.

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Use Cases and Applications

- Connection-oriented protocols shine in scenarios where accuracy is paramount, and the data must arrive in the correct order. Examples include:
 - Web Browsing (HTTP/HTTPS): Ensures websites load completely and accurately.
 - Email (SMTP, POP3, IMAP): Guarantees complete and secure email transmission.
 - File Transfer (FTP, SFTP): Prevents corruption during large file transfers.
 - Secure Shell (SSH): Maintains reliable communication for remote administration.

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- Connectionless protocols excel in environments requiring speed, even if some data loss is acceptable. Examples include:
 - Streaming Media (UDP): Supports live video or audio streaming, where small data losses are imperceptible.
 - Online Gaming (UDP): Prioritizes real-time responsiveness over flawless data delivery.
 - Voice over IP (VoIP) (UDP): Ensures smooth audio transmissions without interruptions from retransmissions.
 - Domain Name System (DNS) (UDP): Resolves domain names quickly, with minimal overhead.
 - Dynamic Host Configuration Protocol (DHCP) (UDP): Rapidly assigns IP addresses without requiring guaranteed delivery.

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